

公式书

数学基础

$$\text{高斯积分: } \int_{-\infty}^{\infty} x^{2n} e^{-\frac{x^2}{2\sigma^2}} dx = \sqrt{2\pi} \sigma^{2n+1} (2n-1)!!$$

$$\int_0^{\infty} x^{2n} e^{-\frac{x^2}{2\sigma^2}} dx = \sqrt{\frac{\pi}{2}} \sigma^{2n+1} (2n-1)!!$$

$$\int_0^{\infty} x^{2n+1} e^{-\frac{x^2}{2\sigma^2}} dx = \sigma^{2n+2} (2n)!!$$

$$\text{伽马函数: } \Gamma(n) = \int_0^{\infty} x^{n-1} e^{-x} dx \quad \Gamma(n+1) = n\Gamma(n) \quad \Gamma(1) = 1 \quad \Gamma(\frac{1}{2}) = \sqrt{\pi}$$

$$\text{R.E. 函数: } g_{\nu}(z) = \frac{1}{\Gamma(\nu)} \int_0^{\infty} \frac{x^{\nu-1}}{e^x - 1} dx = \sum_{n=1}^{\infty} \frac{z^n}{n^{\nu}} = z \frac{d}{dz} g_{\nu+1}(z) \quad g_{\nu}(1) = \zeta(\nu)$$

$$\text{F.D. 函数: } f_{\nu}(z) = \frac{1}{\Gamma(\nu)} \int_0^{\infty} \frac{x^{\nu-1}}{e^x + 1} dx = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{z^n}{n^{\nu}} = z \frac{d}{dz} f_{\nu+1}(z) = -g_{\nu}(-z) \quad f_{\nu}(1) = (1 - \frac{2}{2^{\nu}}) \zeta(\nu)$$

$$\text{黎曼 } \zeta \text{ 函数: } \zeta(\nu) = \sum_{n=1}^{\infty} \frac{1}{n^{\nu}} \quad \zeta(2) = \frac{\pi^2}{6} \quad \zeta(4) = \frac{\pi^4}{90} \quad \zeta(6) = \frac{\pi^6}{945}$$

$$\text{Stirling 公式: } \ln n! \approx n \ln n - n$$

$$\text{d维空间积分: } \int d^d \vec{p} = \frac{2\pi^{d/2}}{\Gamma(\frac{d}{2})} \int_0^{\infty} p^{d-1} dp$$

$$\text{级数求逆: 若 } h = z + a_1 z^2 + a_2 z^3 + a_3 z^4 + \dots$$

$$\text{则 } z = h + (-a_1)h^2 + (2a_1^2 - a_2)h^3 + (-5a_1^3 + 5a_1 a_2 - a_3)h^4 + \dots$$

热力学

$$\text{偏微分变换技巧: ① } \left(\frac{\partial X}{\partial Y} \right)_Z = \frac{1}{\left(\frac{\partial Y}{\partial X} \right)_Z} \quad \left| \frac{\partial(X, Y)}{\partial(Z, W)} \right| = \frac{1}{\left| \frac{\partial(Z, W)}{\partial(X, Y)} \right|}$$

$$\text{② } \left(\frac{\partial X}{\partial Y} \right)_Z = - \left(\frac{\partial X}{\partial Z} \right)_Y \left(\frac{\partial Z}{\partial Y} \right)_X$$

$$\text{③ 令 } F = F(X, Y) = F(Z(X, Y), Y) \quad \left(\frac{\partial F}{\partial X} \right)_Y = \left(\frac{\partial F}{\partial Z} \right)_Y \left(\frac{\partial Z}{\partial X} \right)_Y$$

$$\text{④ 令 } F = F(X, Y) = F(Z(X, Y), X) \quad \left(\frac{\partial F}{\partial X} \right)_Y = - \left(\frac{\partial F}{\partial Z} \right)_X + \left(\frac{\partial F}{\partial Z} \right)_X \left(\frac{\partial Z}{\partial X} \right)_Y$$

$$\text{注: ③④ 可合并为 } dF = \left(\frac{\partial F}{\partial X} \right)_Y dx + \left(\frac{\partial F}{\partial Y} \right)_X dY = \left(\frac{\partial F}{\partial X} \right)_Y \left(\frac{\partial X}{\partial Z} \right)_Y dz + \left[\left(\frac{\partial F}{\partial X} \right)_Y \left(\frac{\partial X}{\partial Y} \right)_Z + \left(\frac{\partial F}{\partial Y} \right)_X \right] dY \\ = \left(\frac{\partial F}{\partial Z} \right)_Y dz + \left(\frac{\partial F}{\partial Y} \right)_Z dY$$

$$\text{热力学第一定律: } dU = \delta Q + \delta W$$

$$\text{热力学第二定律: } dS \geq \frac{\delta Q}{T} \quad \text{卡诺定理: } \eta \leq \eta_{\text{卡诺}} = 1 - \frac{T_1}{T_2}$$

$$\text{热力学基本微分关系与热力学势: } U = U(S, V, N) \quad dU = TdS - pdV + \mu dN$$

$$H = U + pV \quad H = H(S, p, N) \quad dH = TdS + Vdp + \mu dN$$

$$F = U - TS \quad F = F(T, V, N) \quad dF = -SdT - pdV + \mu dN$$

$$G = U + pV - TS \quad G = G(T, p, N) \quad dG = -SdT + Vdp + \mu dN$$

$$\text{迈克耳孙关系: } \left(\frac{\partial T}{\partial V} \right)_S = - \left(\frac{\partial p}{\partial S} \right)_V \quad \left(\frac{\partial T}{\partial p} \right)_S = \left(\frac{\partial V}{\partial S} \right)_p \quad \left(\frac{\partial S}{\partial V} \right)_T = \left(\frac{\partial p}{\partial T} \right)_V \quad \left(\frac{\partial V}{\partial T} \right)_p = - \left(\frac{\partial S}{\partial p} \right)_T$$

$$\text{一般形式: } \left(\frac{\partial f_i}{\partial x_j} \right)_{x_k} = \left(\frac{\partial f_j}{\partial x_i} \right)_{x_k}$$

$$\left(\frac{\partial x_i}{\partial x_j} \right)_{x_k} = - \left(\frac{\partial f_i}{\partial f_j} \right)_{x_k}$$

$$\text{广延量与齐次函数性质: } U = TS - pV + \mu N \quad G(T, p, N) = N\mu(T, p)$$

$$\text{Gibbs-Duhem 关系: } SdT - Vdp + Nd\mu = 0$$



广义力	T	$-P$	μ	σ	$\vec{E}$	$\mu_0 \vec{H}$	ε	f
广义坐标	S	V	N	A	$\vec{p}$	$\vec{m}$	Q	L
微元功	TdS	$-pdV$	μdN	σdA	$\vec{E} \cdot d\vec{p}$	$\mu_0 \vec{H} \cdot d\vec{m}$	εdQ	$f dL$

响应函数: $C_V = \left(\frac{\partial Q}{\partial T}\right)_V = \left(\frac{\partial U}{\partial T}\right)_V = T \left(\frac{\partial S}{\partial T}\right)_V$ $C_P = \left(\frac{\partial Q}{\partial T}\right)_P = \left(\frac{\partial H}{\partial T}\right)_P = T \left(\frac{\partial S}{\partial T}\right)_P$

$C_{H,H} = \left(\frac{\partial Q}{\partial T}\right)_{H,H} = T \left(\frac{\partial S}{\partial T}\right)_{H,H}$ $\mu_H = \left(\frac{\partial T}{\partial P}\right)_H$ J-T系数

$\alpha = \frac{1}{V} \left(\frac{\partial V}{\partial T}\right)_P$ $\beta = \frac{1}{P} \left(\frac{\partial P}{\partial T}\right)_V$ $\kappa_T = -\frac{1}{V} \left(\frac{\partial V}{\partial P}\right)_T$

平衡条件: 可交换广延量 X 则对应广义力 f 相等.

稳定平衡条件: 孤立: $S(U, V)$ 极大 等 T, V ; $F(T, V)$ 极小. 等 T, P ; $G(T, P)$ 极小.

"极大": Hessian 矩阵负定 "极小": Hessian 矩阵正定 等 S, V ; $U(S, V)$ 极小. 等 S, P ; $H(S, P)$ 极小.

统计物理

玻尔兹曼公式: $S = k_B \ln \Omega$

单个自由粒子微观态求和 $\sum_k = \frac{V}{h^3} \int d^3p$ 非相对论: $\varepsilon = \frac{p^2}{2m}$; $\sum_k = \frac{V}{\lambda_{th}^3} \int_0^\infty x^{\frac{1}{2}-1} dx$

$x = \frac{\varepsilon}{k_B T}$; $\lambda_{th} = \frac{h}{\sqrt{2\pi m k_B T}}$

由 N 个自由粒子系统微观态求和 $\sum_{\{k\}} = \frac{V^N}{N! h^{3N}} \int d^3p_1 \dots d^3p_N$ (非定域系统 Gibbs 因子 $\frac{1}{N!}$)

近独立全同粒子的统计分布: 单粒子态 k 的平均占据数 $f(\varepsilon_k)$. $N = \sum_k f_k$. $U = \sum_k f_k \varepsilon_k$

M.B. $f = e^{-\alpha - \beta \varepsilon}$ $\Omega = \frac{N!}{\prod n_k!} \prod \frac{g_k^{n_k}}{n_k!}$ $N = \frac{V}{\lambda_{th}^3} Z$ $U = \frac{1}{2} k_B T \frac{V}{\lambda_{th}^3} Z$ 理想气体

B.E. $f = \frac{1}{e^{\alpha + \beta \varepsilon} - 1}$ $\Omega = \prod \frac{g_k^{n_k} (n_k + W_k - 1)!}{n_k! (W_k - 1)!}$ $N = \frac{V}{\lambda_{th}^3} g_{\frac{1}{2}}(z)$ $U = \frac{1}{2} k_B T \frac{V}{\lambda_{th}^3} g_{\frac{3}{2}}(z)$ 玻色子

F.D. $f = \frac{1}{e^{\alpha + \beta \varepsilon} + 1}$ $\Omega = \prod \frac{g_k^{n_k} W_k!}{n_k! (W_k - n_k)!}$ $N = \frac{V}{\lambda_{th}^3} f_{\frac{1}{2}}(z)$ $U = \frac{1}{2} k_B T \frac{V}{\lambda_{th}^3} f_{\frac{3}{2}}(z)$ 费米子

$\alpha = -\frac{\mu}{k_B T}$ $\beta = \frac{1}{k_B T}$ $z = e^{-\alpha} = e^{\frac{\mu}{k_B T}}$ 逸度

广义力 $f_i = \sum_k f_k \frac{\partial \varepsilon_k}{\partial X_i}$ 如 $-P = \sum_k f_k \frac{\partial \varepsilon_k}{\partial V}$

非相对论气体: $\varepsilon_k \propto V^{-\frac{2}{3}}$ $P = \frac{2}{3} \frac{U}{V}$ 极端相对论气体 $\varepsilon_k = pc \propto V^{-\frac{1}{3}}$ $P = \frac{1}{3} \frac{U}{V}$

注: 对于电子、光子等有内部自由度的在态求和时要 $\times 2$

绝热方程: S 不变 $\Rightarrow n_e$ 不变 $\Rightarrow \beta \varepsilon$ 不变 $\Rightarrow T V^{\frac{2}{3}}$ 不变 (非相对论气体)

能均分定理: 每个 ε 中的 "平方项" 贡献 $\frac{1}{2} k_B T$ 的平均粒子能量

B.E. 凝聚: 条件 $N > \frac{V}{\lambda_{th}^3} g_{\frac{1}{2}}(1)$ 即 $T < T_c$ (T_c 由 $N = \frac{V}{\lambda_{th}^3(T=T_c)} g_{\frac{1}{2}}(1)$ 解得)

$T_c = \frac{h^2}{2\pi m k_B} \left(\frac{N}{V g_{\frac{1}{2}}(1)} \right)^{\frac{2}{3}}$ 此时宏观数量 N_0 的粒子处于最低能级

$T < T_c$ 情况: $N = N_0 + \frac{V}{\lambda_{th}^3(T=T_c)} \zeta(\frac{1}{2}) = \frac{V}{\lambda_{th}^3(T=T_c)} \zeta(\frac{1}{2})$

$\frac{N_0}{N} = 1 - \left(\frac{T}{T_c} \right)^{\frac{3}{2}}$ 注: 非简并玻色气体 $\mu < 0$, 当 $\mu = 0$ 时开始凝聚 deli ("0" 可为 ε 的最小值)



光子气体: N 不守恒, $f = \frac{1}{e^{\frac{\epsilon}{kT}} - 1}$ $\epsilon = pc$ 内简并度 2

弱简并条件: $\mu \ll 0$ $z \ll 1$ $\frac{N\lambda_{th}^d}{V} \ll 1$

零温费米气体: $f = \begin{cases} 1 & \epsilon < \mu \\ 0 & \epsilon > \mu \end{cases}$ $N = \frac{V}{h^d} \frac{(2\pi m)^{\frac{d}{2}}}{\Gamma(\frac{d}{2}+1)} \mu^{\frac{d}{2}}$

费米能 $\epsilon_F = \mu(T=0) = \frac{h^2}{2\pi m} \left(\frac{N\Gamma(\frac{d}{2}+1)}{V} \right)^{\frac{2}{d}}$

费米温度 $T_F = \frac{\epsilon_F}{k_B}$ $\mu = \frac{d}{d+2} N \epsilon_F$

低温费米气体: $T \ll T_F$ $\mu(T) = \mu(T=0) + \frac{1}{2}\pi^2 T^2$

系统理论	微正则系综	正则系综	巨正则系综
固定热力学量	E, N	T, N	T, μ
微观状态和	$\sum_{S_N + N \uparrow \downarrow \text{等} + \text{其他约束}}$	$\sum_{S_N}$	$\sum_{N=0}^{\infty} \sum_{S_N}$
配分函数	$\Omega(E, N, X)$ $= \sum_{S_N} \delta(E - E(S_N, X))$	$Z(T, N, X)$ $= \sum_{S_N} e^{-\beta E(S_N, X)}$ $= \int \Omega(E, N, X) e^{-\beta E} dE$	$\mathcal{Z}(T, \mu, X)$ $= \sum_{N=0}^{\infty} \sum_{S_N} e^{-\beta E} e^{\beta \mu N}$ $= \sum_{N=0}^{\infty} z^N Z(T, N, X) \quad (z = e^{\frac{\mu}{kT}})$
配分函数对应热力学量	$S(E, N, X)$ $= k_B \ln \Omega(E, N, X)$	$F(T, N, X) = U - TS$ $= -k_B T \ln Z(T, N, X)$	$\Phi(T, \mu, X) = U - TS - \mu N$ $= -k_B T \ln \mathcal{Z}(T, \mu, X)$
U, N 的计算	$U = E$	$U = \langle E \rangle = -\frac{\partial}{\partial \beta} (\ln Z)_{N, X}$	$U = \langle E \rangle = -\frac{\partial}{\partial \beta} (\ln \mathcal{Z})_{\mu, X}$ $\langle N \rangle = z \frac{\partial}{\partial z} (\ln \mathcal{Z})_{\mu, X} \quad (z = e^{\frac{\mu}{kT}})$
其他量的计算	$\frac{1}{T} = \frac{\partial S}{\partial E}$ $\frac{p}{T} = \frac{\partial S}{\partial V}$ $\frac{\mu}{T} = -\frac{\partial S}{\partial N}$ $S = -\frac{\partial F}{\partial T}$ $\mu = \frac{\partial F}{\partial N}$ $f_x = \frac{\partial F}{\partial X}$	$S = -\frac{\partial F}{\partial T}$ $\mu = \frac{\partial F}{\partial N}$ $f_x = \frac{\partial F}{\partial X}$	$S = -\frac{\partial \Phi}{\partial T}$ $N = -\frac{\partial \Phi}{\partial \mu}$ $f_x = \frac{\partial \Phi}{\partial X}$
经典粒子简化	无	$Z_1(T, X) = \sum_k e^{-\beta \epsilon_k}$	非空域: $\mathcal{Z} = \exp(z \cdot Z_1)$
(k 为单粒子态)		$Z(T, N, X) = (Z_1)^N$ 空域	$\Phi = -k_B T z \cdot Z_1$
		$Z(T, N, X) = \frac{1}{N!} (Z_1)^N$ 非空域	$\langle N \rangle = z \cdot Z_1$ $U = \langle E \rangle = -z \frac{\partial Z_1}{\partial \beta}$
量子粒子简化	无	无	$\mathcal{Z}_k(T, \mu, X) = \sum_{n_k} e^{\beta(\mu - \epsilon_k) n_k}$
(n_k 为 k 态占据数)			$\mathcal{Z} = \prod_k \mathcal{Z}_k$ $\langle N \rangle = \sum_k \langle n_k \rangle$ $U = \langle E \rangle = \sum_k \langle n_k \rangle \epsilon_k$
			$z \frac{\partial}{\partial z} (\ln \mathcal{Z})_{\mu, X} = \langle n_k \rangle = \frac{1}{e^{\beta(\epsilon_k - \mu)} \mp 1}$ - : 费米 + : 玻色
广义力 $f_x = \langle \frac{\partial E}{\partial X} \rangle$		正则系统涨落公式: $\langle (E - \langle E \rangle)^2 \rangle = k_B T^2 C_V$ $\langle (N - \langle N \rangle)^2 \rangle = k_B T \left(\frac{\partial N}{\partial \mu} \right)_{T, V}$	
如果 $E(S, X) \propto X^\alpha$ 则 $\frac{\partial E(S, X)}{\partial X} = \alpha \frac{E(S, X)}{X}$, $f_x = \alpha \frac{U}{X}$			
补充: 系综密度	$\delta(E - E(S_N, X))$	$e^{-\beta E}$	$e^{-\beta E + \beta \mu N}$
$p(S_N)$	Ω	Z	$\mathcal{Z}$

